Xiaowu Wu

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Summary

Human-Centred Data Science graduate student at the University of Toronto with an interdisciplinary background in psychology, education, and computer science. Experienced in Python, quantitative research, data analysis, and translating complex information into practical, user-centred solutions. Interested in applying analytical problem-solving skills to strategy, customer insights, and data-informed decision-making.

Education

University of Toronto — Toronto, ON
Master of Information, Human-Centred Data Science (Co-op) — *Expected 2028*

University of Toronto — Toronto, ON
Bachelor of Science in Psychology; Minors in Computer Science and Education & Society — *Sep. 2022 – May 2026*

- CGPA: 3.53/4.00 (High Distinction)

Skills

Programming & Data: Python, SQL, PyTorch, pandas, NumPy, scikit-learn, DuckDB, Excel, Git
Research & Analytics: Quantitative and qualitative data analysis, experimental design, hypothesis testing, predictive modeling, model evaluation
Professional: Analytical problem solving, written communication, stakeholder collaboration, presentation, adaptability

Projects

Dota 2 Draft Prediction | *Python, PyTorch, SQL, DuckDB* — 2026 – Present

- Built an end-to-end data pipeline for one year of professional Dota 2 matches using the OpenDota API, normalizing raw JSON into structured match, player, and ban/pick tables with DuckDB, SQL, PyArrow, and Parquet.
- Framed each draft action as a supervised next-hero prediction task, using time-based 70/15/15 train-validation-test splits and only prior ban/pick history; implemented PyTorch MLP and Transformer models with patch, team, and player embeddings.
- Benchmarked model variants against statistical baselines using ranked prediction metrics, iteratively improving prediction accuracy to 31% versus a 20% statistical baseline while preserving legal-candidate constraints.

Industrial Electricity Load Analysis | *Python, pandas, scikit-learn, NumPy* — 2026

- For Shanghai Jushu Technology Co., Ltd., analyzed approximately 2,376 enterprise-day electricity records from four enterprises spanning Sep. 2023–May 2025, normalizing 24-hour load profiles by daily consumption to isolate usage-pattern shape from total demand.
- Built a K-Means segmentation pipeline that evaluated $K = 2$–10 with SSE/elbow diagnostics and compared pooled with company-specific clusters; produced cluster assignments, centroid and mean load curves, PCA visualizations, and data-quality exclusion logs to surface recurring operating patterns.

Beamer+ Interactive Presentation Platform | *JavaScript, HTML/CSS, Flask, Socket.IO* | GitHub — 2026

- Authored a merged feature contribution implementing stateful annotation editing with canvas snapshot history, synchronized undo/redo controls, and heuristic shape recognition for lines, circles, rectangles, and triangles.
- Developed a real-time *Quick Question* audience-response workflow with presenter/student modes, QR-based joining, Socket.IO start/stop events, countdown timers, and live response aggregation across configurable questions.

Experience

Psychology Intern — May 2025 – Sep. 2025
Jingshan Middle School — *Yancheng, China*

- Conducted school-based psychological research examining relationships between students' internet use and academic performance, applying quantitative research methods to investigate behavioural and educational trends.
- Translated psychological concepts into student-facing newsletters and campus activities, communicating complex information in accessible formats for a broad audience.

Educational Intern — Jan. 2025 – May 2025
United for Literacy — *Toronto, ON*

- Collaborated with supervisors and volunteers to implement a feedback-driven tracking system for learner progress and participation, supporting identification of engagement patterns and instructional planning.
- Designed individualized learning resources and digital materials based on learner needs and feedback, improving accessibility and engagement for students from diverse backgrounds.